

EMILY HUANG

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Technical Skills

- **Programming Languages:** HTML/CSS, JavaScript/TypeScript, Java, C/C++, Python, Lua, SQL
- **Frameworks:** React, SpringBoot, ThymeLeaf, Flask
- **Developer Tools:** GitHub, VScode, IntelliJ, PyCharm, Unity, Blender
- **Graphics Editors:** Figma, InkScape, Canva, Aseprite, Piskel

Transferrable Skills

- **Collaborative:** Attentive, Friendly, Cooperative
- **Organization:** Google Calendar, Slack

Technical Projects

- Personal Website** | *HTML, CSS, JavaScript, Piskel, Aseprite, Figma* April 2025 - Present
Burnaby, BC
- Built an interactive HTML/CSS/JavaScript website with custom pixel-art sprites, Figma SVG assets, slideshow functionality, and mouse-tracking effects.
- Brain Tumor Segmentation U-Net Model** | *PyTorch, Tkinter* February - April 2026
Special Topics in Artificial Intelligence (CMPT 419), SFU *Burnaby, BC*
- Built a brain tumor segmenter using PyTorch and a U-Net architecture, integrating it with a Tkinter-based GUI to help users visualize model predictions.
- Fine-Tuning Diffusion Model** | *PyTorch* March - April 2026
Deep Learning (CMPT 420), SFU *Burnaby, BC*
- Fine-tuned an SDXL diffusion model in PyTorch with LoRA weights to improve depth perception and perspective accuracy in AI-generated images, producing more realistic and compositionally aesthetic outputs.
 - Developed a Gradio-based interface with a Python backend to support interactive inference and model testing.
- Emergency Room Wait Time Prediction (First Prize)** | *Scikit-learn, Pandas* February 2026
VanML Hacks 2026, SFU *Burnaby, BC*
- Preprocessed real-world data using pandas, transforming input data into a dataset ready for analysis and modeling.
 - Engineered features and applied scikit-learn to train and optimize a LightGBM model for predicting emergency department wait times.
 - Built a forecasting solution that contributed to a hackathon-winning project, demonstrating strong skills in data preparation, machine learning, and problem-solving.
- Ducks in a Row (Multiplayer Game)** | *Unity, Piskel* October 2025
Simon StormHacks 2025, Simon Fraser University *Burnaby, BC*
- Built a 2D platform game in Unity, focusing on player interactivity and cooperative mechanics, delivering an engaging and teamwork-based gameplay experience.
 - Designed custom assets in Piskel and integrated them into Unity, enhancing visual consistency and quality of gameplay.
- Miniature Golf** | *Love2D, Lua* July - August 2025
Data Communications and Networking (CMPT371), SFU *Burnaby, BC*
- Developed a multiplayer miniature golf game using Love2D/Lua.
 - Built a client-server architecture with TCP sockets and JSON messages to synchronize gameplay across devices.
 - Implemented server-side game logic including physics simulation, ball movement, scoring, goal detection, player connection handling, and real-time state broadcasting.
 - Designed client-side systems for input handling, shoot command transmission, and applying server state updates.
- Sheep Tossing Minigame** | *Unity, Blender* March 2025
DreamHacks 2025, Simon Fraser University *Burnaby, BC*
- Designed and developed a 3D game in Unity to help users relax and fall asleep, focusing on soothing visuals and interactive environments.
 - Modeled and implemented custom 3D assets in Blender, improving the game's visuals and atmosphere.
 - Collaborated in a fast-paced hackathon setting to deliver a functional prototype within a limited timeframe.

Other Work Experience

Tutor, Allset Learning

Nov 2022 - Present

- Applied strong communication and creativity skills to design engaging educational worksheets across subjects.
- Used clear instructional and interpersonal skills to teach Math, Social Studies, and English to children aged 5 to 14.
- Demonstrated leadership, collaboration, and classroom management by leading group classes, encouraging participation, and maintaining a positive learning environment.

Data Entry Assistant, De Yuan Stainless Steel Fabrications Ltd

Aug 2021 - Present

- Organized financial documents and entered receipt data into bookkeeping software

Education

Simon Fraser University, Burnaby, B.C.

September 2022 – Present

- B.Sc. Computing Science